HH-I-1252B
August 18, 1976
SUPERSEDING
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March 4, 1971

FEDERAL SPECIFICATION

INSULATION, THERMAL, REFLECTIVE (ALUMINUM FOIL)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers aluminum foil insulation.

1.2 Classification.

1.2.1 Forms and classes. The insulation shall be of the following forms and classes as specified (see 6.2).

1.2.1.1 Form 1. Materials providing a minimum 19. millimeters (mm) (3/4-inch) reflective air space having an effective emittance (E) of 0.05 maximum:

Classes shall define the number of reflective air spaces assuming the material is correctly installed.

Class I - One reflective air space.
Class II - Two reflective air spaces.
Class III - Three reflective air spaces.
Class IV - Four reflective air spaces.

1.2.1.2 Form 2. Materials providing a minimum 10. mm (3/8-inch) reflective air space having an effective E of 0.05 maximum:

Class V - One or more reflective air spaces in masonry wall structure.

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2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-B-640 - Boxes, Fiberboard Corrugated, Triple Wall.

Federal Standard:

Fed. Std. No. 123 - Marking for Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards, required by contractors in connection with specific procurement functions, should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM) Standards:

C236 - Thermal Conductance and Transmittance of Built-up Sections by Means of the Guarded Hot Box, Test for.
C445 - Normal Total Emittance of Surfaces of Materials 0.01 Inch or Less in Thickness at Approximately Room Temperature.
D585 - Sampling and Accepting a Single Lot of Paper, Paperboard, Fiberboard, or Related Product, Standard Method of.
E96 - Water Vapor Transmission of Materials in Sheet Form.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the National Motor Freight Traffic Association, Inc., Tariff Order Section, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Tariff Publishing Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Standard product. The thermal insulation delivered under this specification shall be of the manufacturer's commercially current standard product, except for any change necessary to comply with specification requirements.

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3.2 Material. All material shall be new and unused. All material shall be as specified herein. Material not specified shall be of the same quality used for the intended purpose in commercial practice. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual component or the overall assembly.

3.2.1 Foil. Foil used in insulation shall contain not less than 99 percent aluminum.

3.2.2 Kraft paper. Kraft paper used shall meet the requirements of ASTM D585.

3.2.2.1 Flangeboard. (A) Multiple insulation layers (more than 2). If single sheet flangeboard is used, it shall be of 28 point grade (minimum). If double sheet flangeboard is used, it shall be of 14 point grade (minimum). (B) Single or double insulation layers. Flangeboard shall be of 14 point grade (minimum). The requirements of ASTM D585 shall apply.

3.2.3 Adhesive. Adhesive used in bonding shall be waterproof material and shall show no sign of bleeding when tested in accordance with 4.4.5. Bleeding at cut edges may be disregarded.

3.3 Finished insulation. Layers shall be securely bonded together along edges, forming an attachment flap suitable for nailing or stapling.

3.3.1 Class I. Insulation shall be designed to be installed so that the reflective surface is parallel to, and a minimum of 19. mm (3/4 inch) from the opposite face of the air space.

3.3.2 Class II. Insulation shall be designed to be installed so that the reflective surfaces are parallel to, and a minimum of 19. mm (3/4 inch) from the opposite faces of the two air spaces.

3.3.3 Class III. Insulation shall be designed to be installed so that full separation of layers occur and the reflective surfaces are parallel to, and a minimum of 19. mm (3/4 inch) from the opposite faces of the three air spaces.

3.3.4 Class IV. Insulation shall be designed to be installed so that full separation of layers occur and the reflective surfaces are parallel to, and a minimum of 19. mm (3/4 inch) from the opposite faces of the four air spaces.

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3.3.5 Class V. Insulation shall be designed to be installed between 25. mm (1-inch) (nominal) furring strips, so that the reflective surfaces are parallel to, and a minimum of 10. mm (3/8 inch) from the opposite faces of the air spaces.

3.4 Size.

3.4.1 Thickness. Layers of insulation composed of unsupported foil that is exposed shall have a minimum thickness of 0.010 mm (0.0004 inch). Unsupported foil that is sandwiched in a multilayer sheet shall have a minimum thickness of 0.009 mm (0.00035 inch). Foil bonded to kraft paper shall have a minimum thickness of 0.00025 inch.

3.4.2 Widths. Insulation shall be furnished in widths that fit framing members set on 305, 406, 508, or 610 mm (12-, 16-, 20-, or 24-inch) centers as specified (see 6.2).

3.5 Emittance. The foil shall have E not in excess of 0.05, when tested in accordance with 4.4.2.

3.6 Water vapor permeability. When tested in accordance with 4.4.3, all classes of insulation shall have a permeability not in excess of 1.0 perms for vapor pressure differential of 25. mm (1 inch) of mercury. Perm rating of the insulation shall be included on product label.

3.7 Thermal resistance. When determined in accordance with 4.4.6, the thermal resistance (R) value, depending on direction of heat flow, shall comply with the minimum values as shown in Table I.

TABLE I. "R" values (minimum)

Form	Class	"R" values 1/		
		Up	Horizontal	Down
1	I	1.7	2.8	3.0
1	II	3.0	4.7	5.5
1	III	5.7	6.5	8.0
1	IV	7.0	7.8	11.0
1	V	---	3.0	---
2				

1/ "R" values are to be shown for direction of heat flow on package label or product.

3.7.1 "R" value. "R" is the reciprocal of the thermal conductance (C) and shall be calculated as follows: R = 1/C, when C = thermal conductance.

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3.8 Pliability. The finished insulation of all classes shall not crack when folded to 180° bend at a temperature of 21.° Celsius (C) ±1.1°C (70° Fahrenheit (F) ±2°F), and a relative humidity of 50 ±5 percent.

3.9 Workmanship. The insulation shall have uniform flat surfaces and shall not be crumpled, torn, or punctured. Pin point design is an acceptable commercial standard for specific applications.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the contractor may use his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Inspection of materials and components. In accordance with 4.1, the contractor is responsible for insuring that materials and components used were manufactured, examined, and tested to the extent specified, in accordance with the requirements of referenced subsidiary specifications and standards, if any.

4.2 Sampling.

4.2.1 Lot. Unless otherwise specified (see 6.2), a lot shall consist of all rolls of insulation of the same class or classes, and width offered for delivery at one time.

4.2.2 Dimensional and visual examination of end item. Sampling for end item characteristics shall be in accordance with level S-3 in MIL-STD-105.

4.2.3 Sampling for tests. Samples for tests shall be in accordance with level S-2 in MIL-STD-105.

4.3 Examination.

4.3.1 End item. Insulation shall be examined for defects listed in Table II. The Acceptable Quality Level (AQL) shall be 2.5 defects per 100 units for major defects and 4.0 defects per 100 units for minor defects, in accordance with MIL-STD-105.

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TABLE II. Classification of defects.

Defects	Major	Minor
Class not as specified.	X	
Foil thickness less than specified.	X	
Insulation not of correct width as specified.	X	
Attachment edges not of sufficient size or durability to permit nailing or stapling.	X	
Separation of layer edges of multilayer insulation.	X	
Distance of less than 19. mm (3/4 inch) between layers when insulation is expanded.	X	
Torn or punctured.	X	
Crumpled.	X	
Damaged or defects not affecting function or serviceability.		X

4.4 Test methods.

4.4.1 Acceptance. The AQL for tests shall be 6.5 on the basis of defects per 100 units.

4.4.2 Emittance. Each sample selected in accordance with 4.2.3 shall be tested in accordance with ASTM C445.

4.4.3 Water vapor permeability. Each sample selected in accordance with 4.2.3 shall be tested in accordance with procedure A of ASTM E96.

4.4.4 Pliability. Foil shall be folded and the folded edge smoothed using a light finger pressure.

4.4.5 Bleeding adhesive. Specimens for tests shall consist of pieces of insulation cut to approximately 76. by 152. mm (3 by 6 inches) in size. Suspend specimens in vertical position and heat to a temperature of 82.°C ±2.8°C (180°F ±5°F) for at least 5 hours. At the end of heating period, examine the reflective surfaces to determine whether the adhesive has bled or exuded through the surface, or delamination has occurred.

4.4.6 Thermal resistance. "R" of the insulating material only shall be calculated from the thermal conductance tests performed on specimens of production samples of foil, utilizing a guarded hot box and heat flow meter apparatus, or similar apparatus whose calibration is traceable to a guarded hot box apparatus in accordance with ASTM C236. The method used for calculating framing corrections must be stated.

4.5 Inspection of preparation for delivery. The packaging, packing, and marking of the insulation shall be inspected to determine conformance to the applicable requirements of Section 5 of this specification.

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5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A. Each roll shall be packaged in a close-fitting box conforming to PPP-B-636, grade W5c.

5.1.2 Level C. The rolls shall be packaged in accordance with the supplier's standard practice.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A. The rolls of like descriptions shall be packed in boxes conforming to PPP-B-601, overseas type.

5.2.2 Level B. The rolls of like description shall be packed in fiberboard boxes conforming to PPP-B-636, weather-resistant; or PPP-B-640, class 2.

5.2.3 Level C. The insulation shall be packed to insure carrier acceptance and safe delivery to destination at lowest rates in compliance with Uniform Freight Classification rules or National Motor Freight Classification rules.

5.3 Marking. In addition to any special marking required by the contract, interior packages, and shipping containers shall be marked in accordance with Fed. Std. No. 123 or MIL-STD-129, as applicable (see 6.2).

6. NOTES

6.1 Intended use. Foil insulation is intended for insulating floors, walls, and ceilings of buildings. It should be installed as flush or recess installation subject to the manufacturer's recommendation.

6.2 Ordering data. Purchasers should select the perferred options permitted herein and include the following information in procurement documents.

(a) Title, number, and date of this specification.
(b) Form and class desired (see 1.2.1).
(c) Width of insulation desired (see 3.4.2).
(d) Size of lot, if different (see 4.2.1).
(e) Level of packaging and packing required (see 5.1 and 5.2).
(f) Marking required (see 5.3).
(g) Quantity (area) of insulation required (see 6.3).

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6.3 Insulation. Insulation should cover the required area, as specified (see 6.2). Length of a particular roll of the material is not a critical factor, since this type insulation is commonly packaged in continuous lengths up to several hundred meters.

MILITARY CUSTODIANS:

Navy - YD
Air Force - 84

User activities:

Army - CE
Navy - SH

Civil Agency Coordinating Activity:

GSA-FSS

Preparing activity:

Navy - YD

Project No. 5640-0296

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